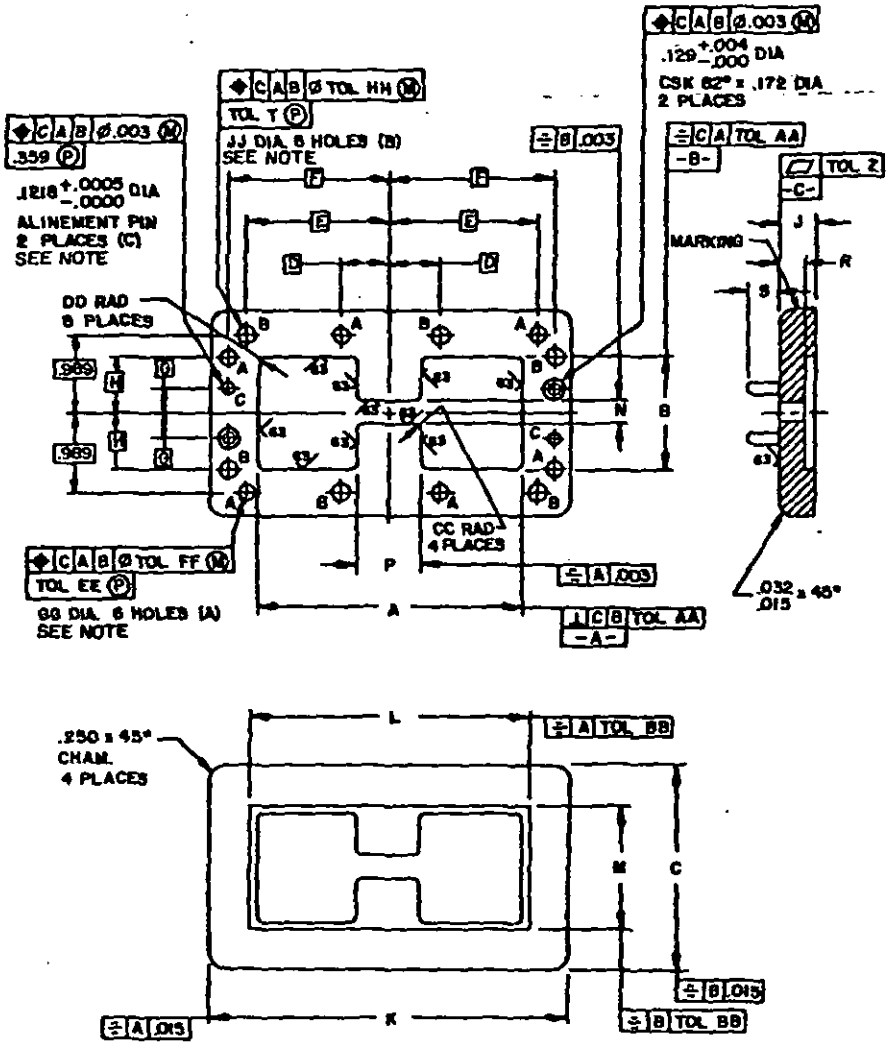
MILITARY SPECIFICATION SHEET

FLANGES, WAVEGUIDE, DOUBLE RIDGE, SOCKET MOUNT (BANDWIDTH RATIO 3.6:1)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the flanges described herein shall consist of this document and the latest issue of Specification MIL-F-39000.

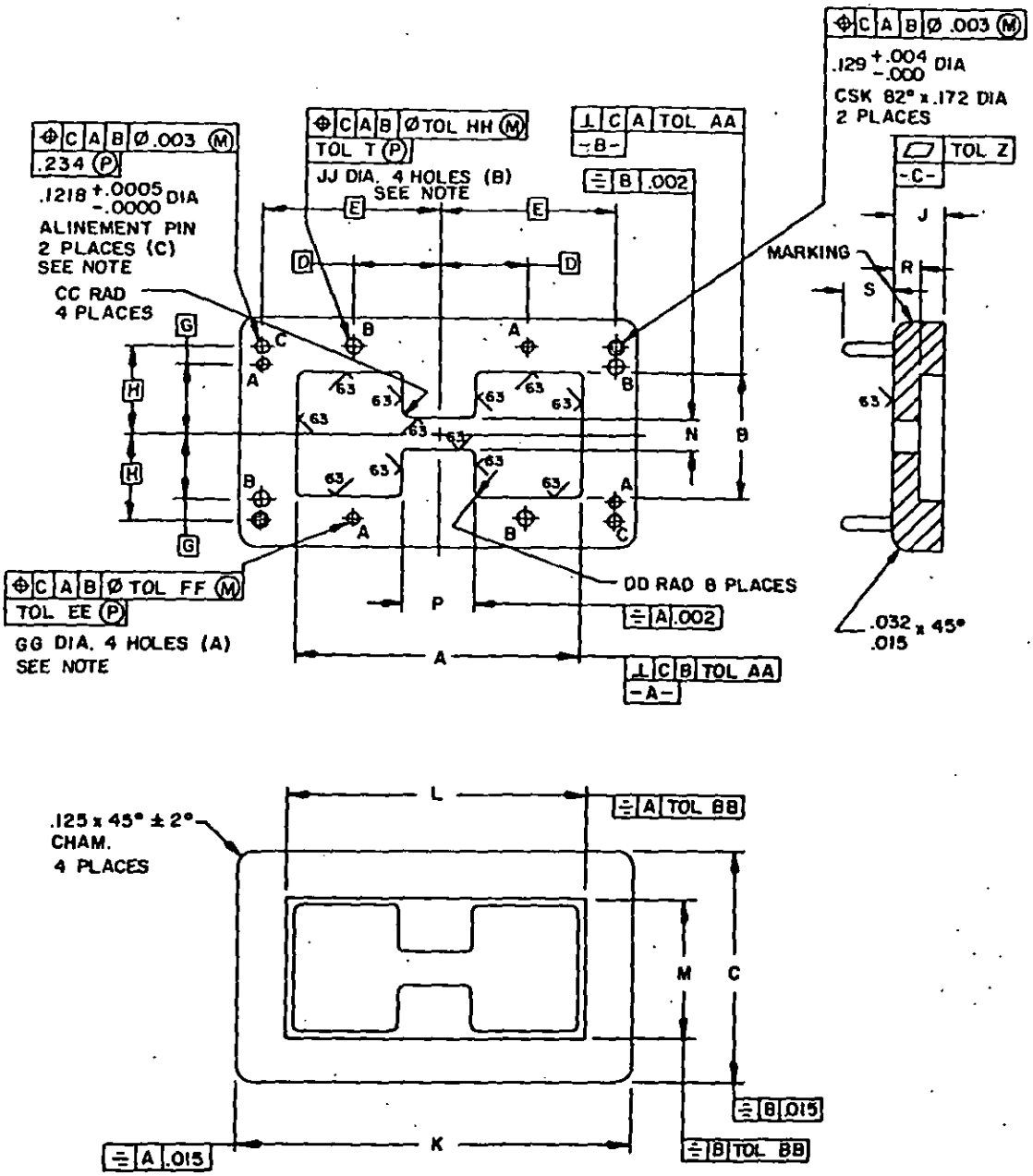


Configuration 1.

FIGURE 1. Flange.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, Department of the Navy, ATTN: ELEX 5043, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

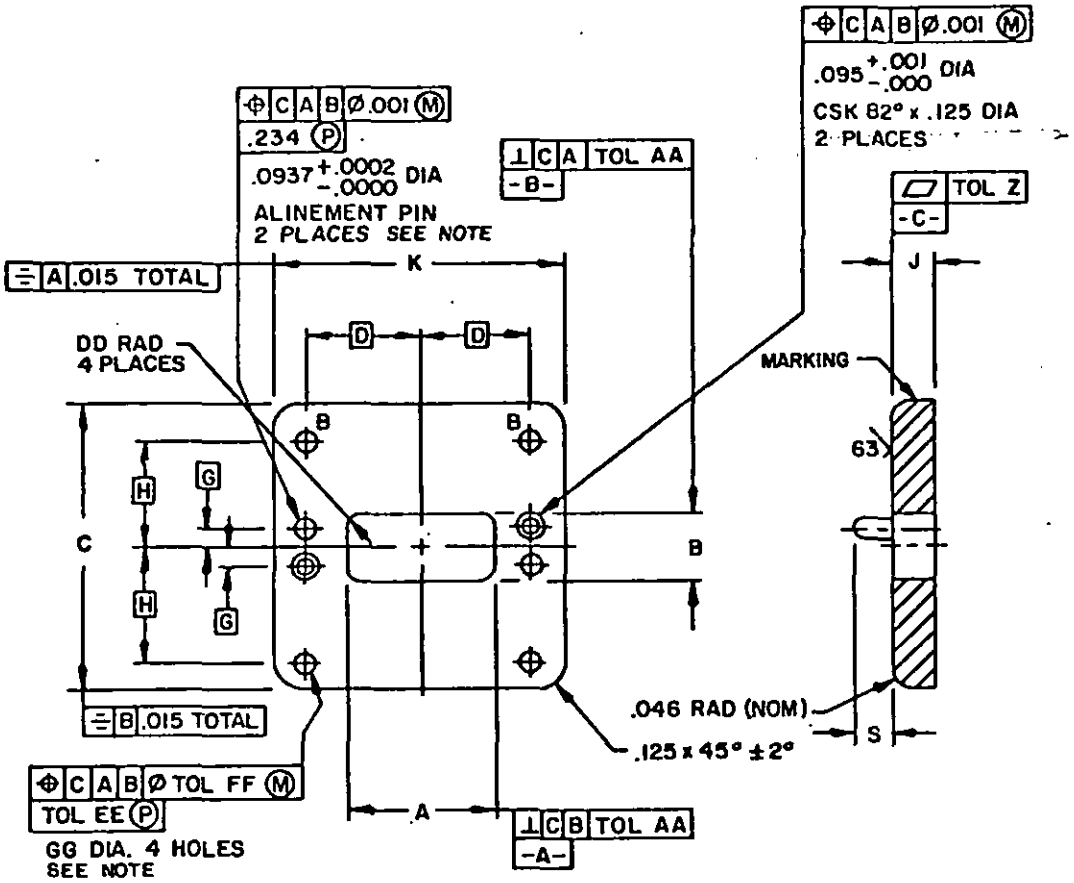
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Configuration 2.

FIGURE 1. Flange - Continued.

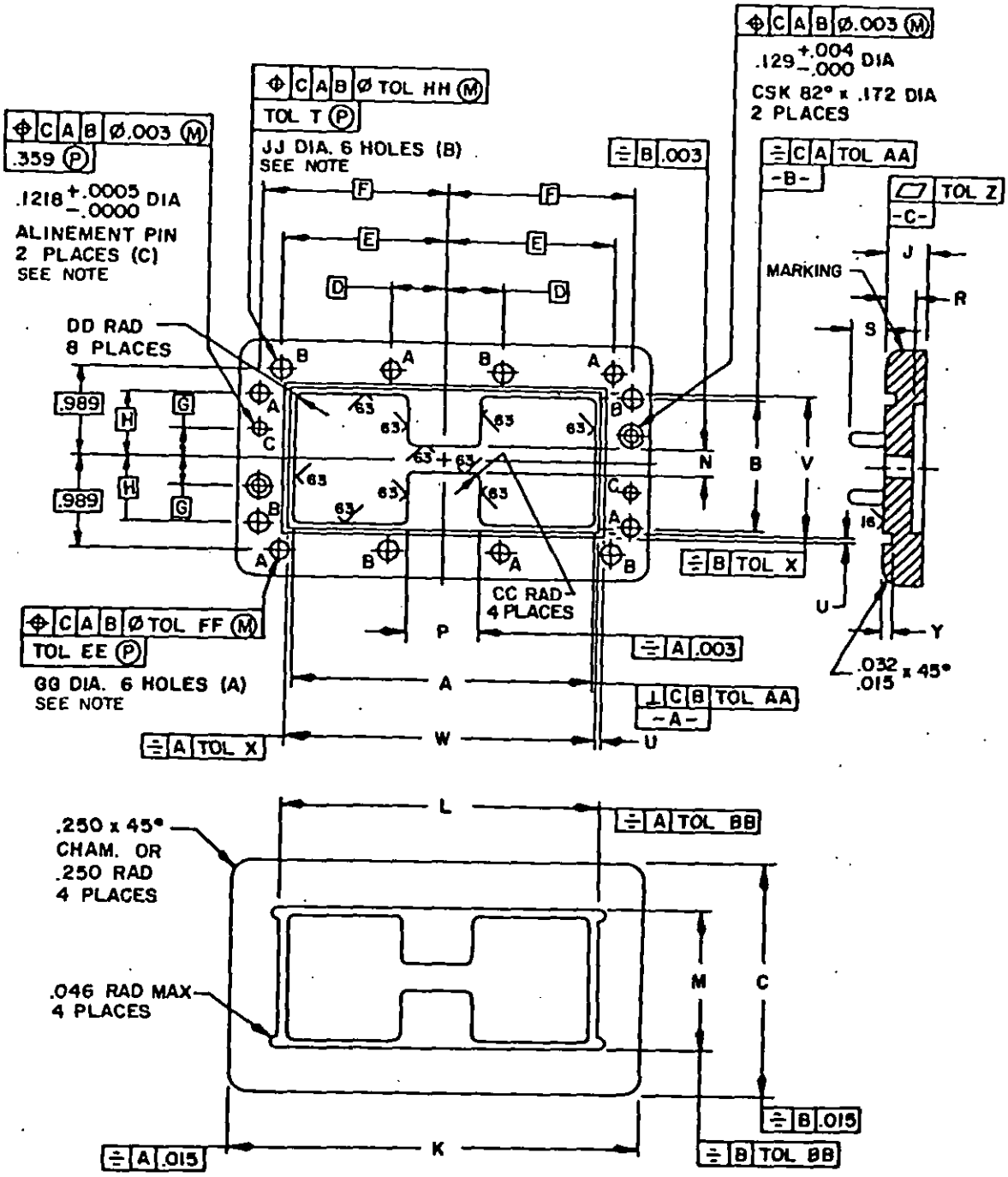
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Configuration 3.

FIGURE 1. Flange - Continued.

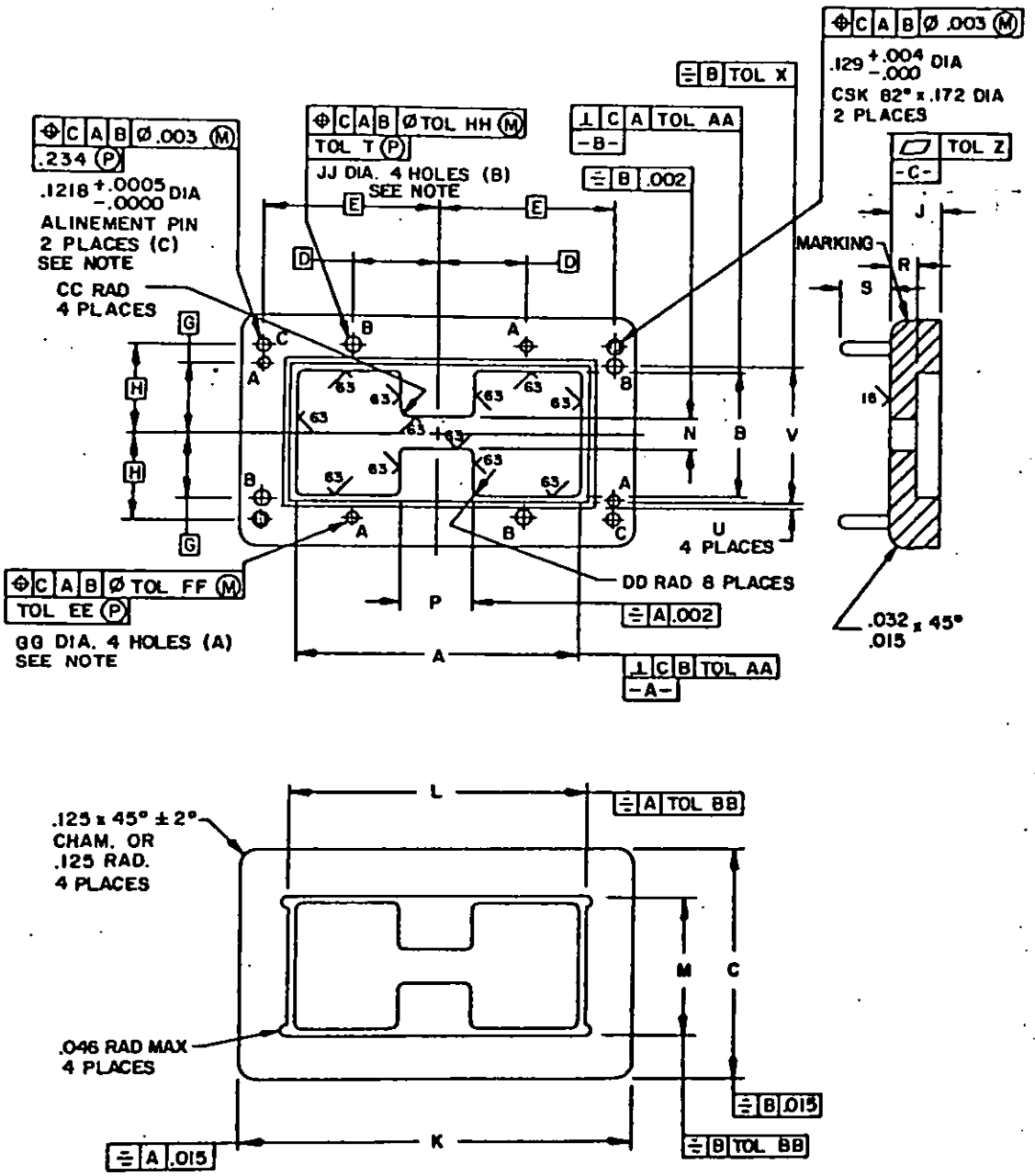
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Configuration 4.

FIGURE 1. Flange - Continued.

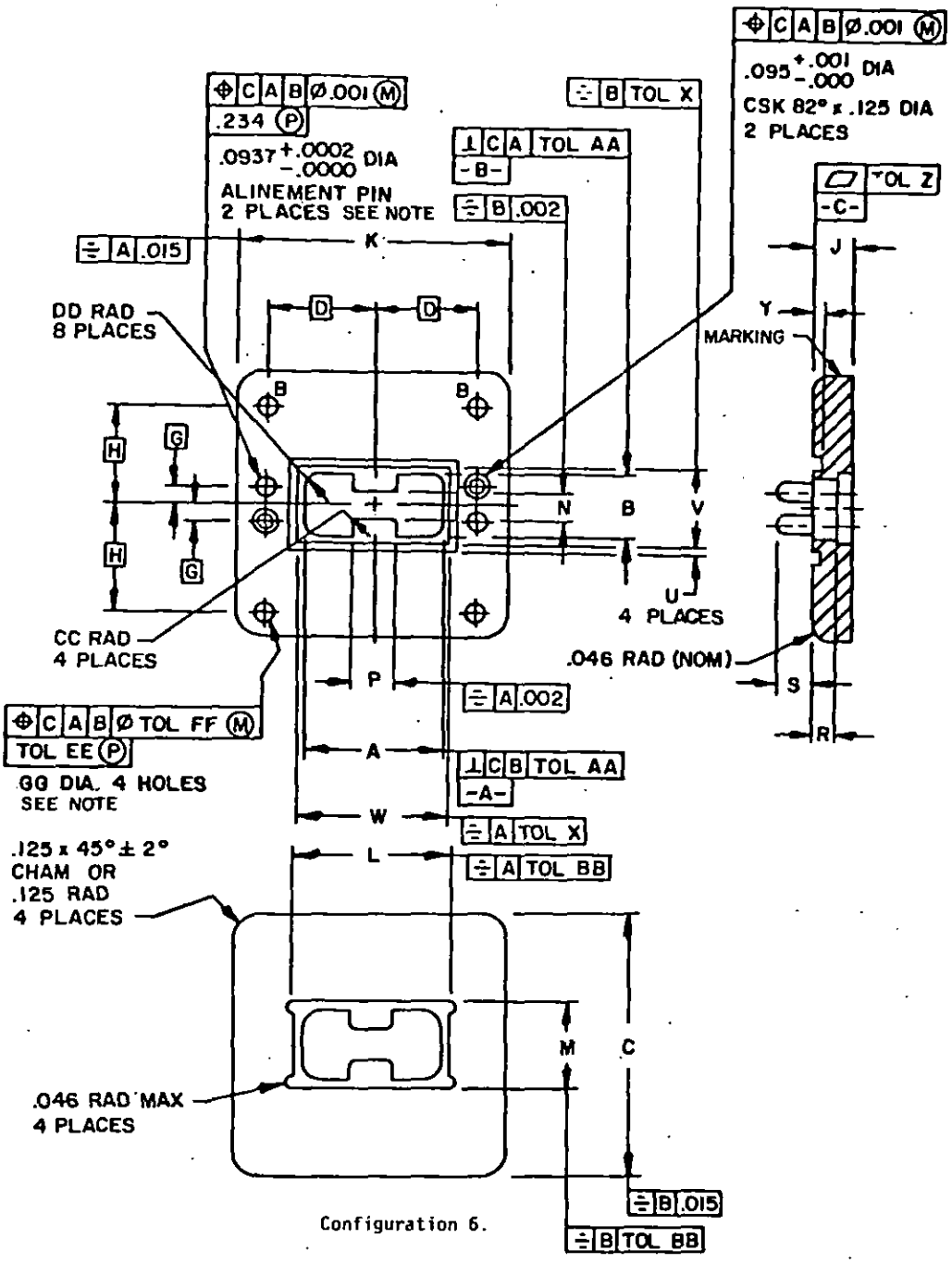
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Configuration 5.

FIGURE 1. Flange - Continued.

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Configuration 6.

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.0002	.005	.004	.10	.095	2.41	.234	5.94
.0005	.013	.015	.38	.1218	3.094	.250	6.35
.001	.02	.032	.81	.125	3.18	.359	9.12
.002	.05	.046	1.17	.129	3.28	.989	25.12
.003	.08	.0937	2.380	.172	4.37		

NOTE: The projected tolerance zone surface reference is datum C.

FIGURE 1. Flange - Continued.

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TABLE I. Part number dimensions and configuration.

Part number M39000/4-			Part number M39000/4-			Config-uration	Dimensions 1/ 2/ 3/	
	Used with			Used with				
Dash No. 4/	Waveguide M23351/2-	Mating flange M39000/4-	Dash No. 5/	Waveguide M23351/2-	Mating flange M39000/4-		A	B
001	013	001, 016	002	014, 015, 016	002, 017	1	3.877 ±.004 (98.48) (.10)	1.667 ±.004 (42.34) (.10)
004	017	004, 018	005	018, 019, 020	005, 019	2	2.687 ±.004 (68.25) (.10)	1.155 ±.004 (29.34) (.10)
007	021	007, 020	008	022, 023, 024	008, 021	2	1.074 ±.003 (27.28) (.08)	.462 ±.003 (11.73) (.08)
010	025	010, 022	011	026, 027, 028	011, 023	3	.859 ±.002 (21.82) (.05)	.430 ±.002 (10.92) (.05)
013	029	013, 024	014	030, 031, 032	014, 025	3	.382 ±.002 (9.70) (.05)	.216 ±.002 (5.49) (.05)
016	013	001, 016	017	014, 015, 016	002, 017	4	3.877 ±.004 (98.48) (.10)	1.667 ±.004 (42.34) (.10)
018	017	004, 018	019	018, 019, 020	005, 019	5	2.687 ±.004 (68.25) (.10)	1.155 ±.004 (29.34) (.10)
020	021	007, 020	021	022, 023, 024	008, 021	5	1.074 ±.003 (27.28) (.08)	.462 ±.003 (11.73) (.08)
022	025	010, 022	023	026, 027, 028	011, 023	6	.752 ±.003 (19.10) (.08)	.323 ±.002 (8.20) (.05)

Dash No. 4/	Dash No. 5/	Dimensions 1/ 2/ 3/								
		C ±.015 (.38)	D BSC.	E BSC.	F BSC.	G BSC.	H BSC.	J ±.015 (.38)	K ±.015 (.38)	L
001	002	2.552 (64.82)	.711 (18.06)	2.140 (54.36)	2.390 (60.71)	.125 (3.18)	.925 (23.50)	.500 (12.70)	5.281 (134.14)	4.048 ±.005 (102.82) (.13)
004	005	2.156 (54.76)	.828 (21.03)	1.656 (42.06)	---	.577 (14.66)	.828 (21.03)	.438 (11.13)	3.687 (93.65)	2.855 ±.003 (72.52) (.08)
007	008	1.312 (33.32)	.457 (11.61)	.832 (21.13)	---	.107 (2.72)	.469 (11.91)	.250 (6.35)	1.938 (49.23)	1.182 ±.003 (30.02) (.08)
010	011	1.250 (31.75)	.635 (16.13)	---	---	.100 (2.54)	.415 (10.54)	.250 (6.35)	1.688 (42.88)	---
013	014	1.000 (25.40)	.303 (7.70)	---	---	.100 (2.54)	.216 (5.49)	.125 (3.18)	1.188 (30.18)	---
016	017	2.552 (64.82)	.711 (18.06)	2.140 (54.36)	2.390 (60.71)	.125 (3.18)	.925 (23.50)	.500 (12.70)	5.281 (134.14)	4.048 ±.005 (102.82) (.13)
018	019	2.156 (54.76)	.828 (21.03)	1.656 (42.06)	---	.577 (14.66)	.828 (21.03)	.438 (11.13)	3.687 (93.65)	2.855 ±.003 (72.52) (.08)
020	021	1.312 (33.32)	.457 (11.61)	.832 (21.13)	---	.107 (2.72)	.469 (11.91)	.250 (6.35)	1.938 (49.23)	1.182 ±.003 (30.02) (.08)
022	023	1.250 (31.75)	.635 (16.13)	---	---	.100 (2.54)	.415 (10.54)	.250 (6.35)	1.688 (42.88)	.860 ±.003 (21.84) (.08)

See footnotes at end of table.

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TABLE I. Part number dimensions and configuration - Continued.

Dash No. 4/	Dash No. 5/	Dimensions 1/ 2/ 3/					
		M	N	P	R ±.005 (.13)	S ±.005 (.13)	T
001	002	1.838 ±.005 (46.69) (.13)	.325 ±.003 (8.26) (.08)	.969 ±.003 (24.61) (.08)	.125 (3.18)	.359 (9.12)	---
004	005	1.323 ±.003 (33.60) (.08)	.225 ±.002 (5.72) (.05)	.672 ±.002 (17.07) (.05)	.188 (4.78)	.422 (10.72)	---
007	008	.570 ±.003 (14.48) (.08)	.090 ±.002 (2.29) (.05)	.269 ±.002 (6.83) (.05)	.125 (3.18)	.234 (5.94)	---
010	011	---	---	---	---	.234 (5.94)	---
013	014	---	---	---	---	.109 (2.77)	---
016	017	1.838 ±.005 (46.69) (.13)	.325 ±.003 (8.26) (.08)	.969 ±.003 (24.61) (.08)	.125 (3.18)	.359 (9.12)	---
018	019	1.323 ±.003 (33.60) (.08)	.225 ±.002 (5.72) (.05)	.672 ±.002 (17.07) (.05)	.188 (4.78)	.422 (10.72)	---
020	021	.570 ±.003 (14.48) (.08)	.090 ±.002 (2.29) (.05)	.269 ±.002 (6.83) (.05)	.125 (3.18)	.234 (5.94)	---
022	023	.431 ±.003 (10.95) (.08)	.063 ±.002 (1.60) (.05)	.188 ±.002 (4.78) (.05)	.125 (3.18)	.234 (5.94)	---

Dash No. 4/	Dash No. 5/	Dimensions 1/ 2/ 3/					
		U	V	W	X	Y	Z
001	002	---	---	---	---	---	.0007 (.018)
004	005	---	---	---	---	---	.0007 (.018)
007	008	---	---	---	---	---	.0004 (.010)
010	011	---	---	---	---	---	.0003 (.008)
013	014	---	---	---	---	---	.0003 (.008)
016	017	.094 ±.002 (2.39) (.05)	1.787 ±.004 (45.39) (.10)	3.997 ±.004 (101.52) (.10)	---	.056 ±.002 (1.42) (.05)	.0007 (.018)
018	019	.094 ±.002 (2.39) (.05)	1.275 ±.004 (32.38) (.10)	2.807 ±.004 (71.30) (.10)	---	.056 ±.002 (1.42) (.05)	.0007 (.018)
020	021	.094 ±.002 (2.39) (.05)	.582 ±.004 (14.78) (.10)	1.194 ±.004 (30.33) (.10)	---	.056 ±.002 (1.42) (.05)	.0004 (.010)
022	023	.094 ±.002 (2.39) (.05)	.443 ±.004 (11.25) (.10)	.872 ±.004 (22.15) (.10)	---	.056 ±.002 (1.42) (.05)	.0003 (.008)

See footnotes at end of table.

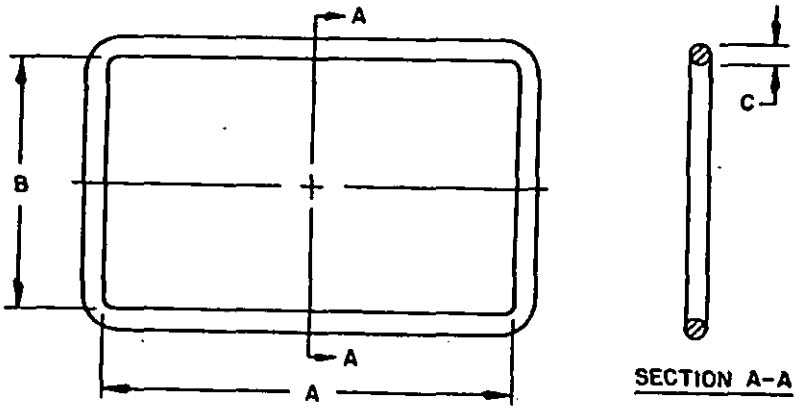
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TABLE I. Part number dimensions and configuration - Continued.

Dash No. 4/	Dash No. 5/	Dimensions 1/ 2/ 3/								
		AA	BB	CC ±10%	DD Max.	EE	FF	GG	HH	JJ
001	002	.004 (.10)	.005 (.13)	.065 (1.65)	.050 (1.27)	.500 (12.70)	.005 (.13)	.250-20 UNC-2B (6.35)	.004 (.10)	.264 ±.005 (6.71) (.13)
004	005	.004 (.10)	.003 (.08)	.045 (1.14)	.050 (1.27)	.438 (11.13)	.004 (.10)	.138-32 UNC-2B (3.50)	.004 (.10)	.148 ±.002 (3.76) (.05)
007	008	.003 (.08)	.003 (.08)	.018 (.46)	.030 (.76)	.250 (6.35)	.004 (.10)	.138-32 UNC-2B (3.50)	.004 (.10)	.148 ±.002 (3.76) (.05)
010	011	.002 (.05)	---	---	.075 (1.90)	---	.007 (.18)	.128 .001 (3.25) (.03)	---	---
013	014	.002 (.05)	---	---	.060 (1.52)	---	.007 (.18)	.128 .001 (3.25) (.03)	---	---
016	017	.004 (.10)	.005 (.13)	.065 (1.65)	.050 (1.27)	.500 (12.70)	.005 (.13)	.250-20 UNC-2B (6.35)	.004 (.10)	.264 ±.005 (6.71) (.13)
018	019	.004 (.10)	.003 (.08)	.045 (1.14)	.050 (1.27)	.438 (11.13)	.004 (.10)	.138-32 UNC-2B (3.50)	.004 (.10)	.148 ±.002 (3.76) (.05)
020	021	.003 (.08)	.003 (.08)	.018 (.46)	.030 (.76)	.250 (6.35)	.004 (.10)	.138-32 UNC-2B (3.50)	.004 (.10)	.148 ±.002 (3.76) (.05)
022	023	.002 (.05)	.003 (.08)	---	.075 (1.90)	---	.007 (.18)	.128 .001 (3.25) (.03)	---	---

1/ Dimensions are in inches.
2/ Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3/ Metric equivalents are in parentheses.
4/ Material shall be aluminum alloy.
5/ Material shall be copper alloy.

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Used with part no.	A	B	C
016, 017	3.997 ±.005 (101.52) (.13)	1.787 ±.005 (45.39) (.13)	.070 ±.003 (1.78) (.08)
018, 019	2.807 ±.005 (71.30) (.13)	1.275 ±.005 (32.38) (.13)	.070 ±.003 (1.78) (.08)
020, 021	1.194 ±.005 (30.33) (.13)	.582 ±.005 (14.78) (.13)	.070 ±.003 (1.78) (.08)
022, 023	.872 ±.005 (22.15) (.13)	.443 ±.005 (11.25) (.13)	.070 ±.003 (1.78) (.08)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Metric equivalents are in parentheses.

FIGURE 2. Gasket.

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REQUIREMENTS:

Dimensions and configuration:
Flange: See figure 1 and table I.
Gasket: See figure 2.

Material: Metallic alloy as specified in table I.
Materials supplied with flanges: See table II.

Marking:
Part number: M39000/4-(dash number from table I).
Location: See figure 1.

Cross reference and frequency range: See table III.

TABLE II. Items supplied with flanges.

Flange part No. M39000/4-	Socket head cap screw	Hex nut	Spring lockwasher	Gasket
001, 002	.250-20 UNC-2A 1.5 in. long (6 ea)	---	.250 x .062 (6 ea)	---
004, 005	.138-32 UNC-2A 1.5 in. long (4 ea)	---	.138 x .031 (4 ea)	---
007, 008	.138-32 UNC-2A 0.75 in. long (4 ea)	---	.138 x .031 (4 ea)	---
010, 011	.112-40 UNC-2A 0.75 in. long (2 ea)	.112-40 UNC-2B (2 ea)	.112 x .025 (2 ea)	---
013, 014	.112-40 UNC-2A 0.5 in. long (2 ea)	.112-40 UNC-2B (2 ea)	.112 x .025 (2 ea)	---
016, 017	.250-20 UNC-2A 1.5 in. long (6 ea)	---	.250 x .062 (6 ea)	See figure 2
018, 019	.138-32 UNC-2A 1.5 in. long (4 ea)	---	.138 x .031 (4 ea)	See figure 2
020, 021	.138-32 UNC-2A 0.75 in. long (4 ea)	---	.138 x .031 (4 ea)	See figure 2
022, 023	.112-40 UNC-2A 0.75 in. long (2 ea)	.112-40 UNC-2B (2 ea)	.112 x .025 (2 ea)	See figure 2

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TABLE III. Cross reference and frequency range.

Part number 1/ M39000/4-	AN nomenclature	Frequency range (GHz)
001	UG-1589/U	.97- 3.50
002	UG-1590/U	.97- 3.50
004	UG-1592/U	1.40- 5.00
005	UG-1593/U	1.40- 5.00
007	UG-1595/U	3.50-12.40
008	UG-1596/U	3.50-12.40
010	UG-1598/U	5.00-18.00
011	UG-1599/U	5.00-18.00
013	UG-1601/U	12.40-40.00
014	UG-1602/U	12.40-40.00
016, 017	—	.97- 3.50
018, 019	—	1.40- 5.00
020, 021	—	3.50-12.40
022, 023	—	5.00-18.00

1/ Dash numbers 003 (UG-1591/U), 006 (UG-1594/U), 009 (UG-1597/U), 012 (UG-1600/U), and 015 (UG-1603/U) have been deleted from specifications.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army -
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Army - AV, AR
Navy - MC, AS, OS, SH, CG
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5985-0854-4)

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